Trading with Multiple Monitors

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Part 1 – How Multi-Screen Trading Computers Actually Work

Technically a multi-screen capable trading computer isn't too dissimilar to a standard computer, it uses the same type of components such as a central processor, motherboard, hard drive, etc, what is different though are the graphics cards.

Graphics cards in a computer perform two jobs.

1. They perform the data processing that turns bits of information, the 1's and 0's, into graphical images that you and I understand.
2. They physically allow for connection of a screen to the PC which can then display the output of the computer.

Back when personal / home computers were first becoming more common, a graphics card would have just one monitor port allowing for the connection of one screen.

There was no way to connect up a second screen, and even if you could have the software at the time would not have been able to make use of the extra screen.

Fast forward a few years and the widespread adoption of computers to handle pretty much any task led to big improvements in hardware performance levels and software package capabilities.

Now a days running a couple of screens whilst still not standard is a lot more common than it ever has been, most graphics cards can handle at least two simultaneous monitor connections.

So, if you want to run a computer with 4 screens connected then you need a graphics card setup inside the PC that can process the output for each of the screens, and then can physically connect up to four separate screens.

There are a few ways of achieving this however if you want a stable, responsive and really quiet trading computer then the best way is to use multiple graphics cards.

Note: We go into the different types of graphics card setups in a lot of detail here if you are interested in reading further: www.multiplemonitors.co.uk/blog/what-you-need-to-know-about-multi-monitor-graphics-cards/

That's how we achieve a multi-screen system but is that the only thing you need to be aware of when finding a new trading computer? We don't think so.

There are many other details which go into making a fantastic trading system including:

- System speed and responsiveness so you can both see and react to opportunities instantly
- System stability, being able to handle long trading sessions and prolonged usage

Part 1 – How Multi-Screen Trading Computers Actually Work Continued...

- The ability to quickly recover from system errors
- Noise levels (or making sure you're not sat next to headache inducing noise all day long)
- Software setup to ensure that your system is helping you make the best use of your hardware

Many of these factors are often overlooked when people are deciding on a system to support their trading, however we believe they are all important aspects of a trading computer build, and they will lead to a PC that will become an important part of your future trading success.

Using a Multi-Screen Computer

We have briefly covered what is technically required for a new trading computer system build however once the machine is built and you've got your screens connected how do you actually use it?

This is a question we get asked a lot and if you have never used a multi-screen PC before then it can be difficult to understand exactly how one would work on a day to day basis.

Essentially I ask customers to picture their current computer (or laptop) with one screen running, now imagine that a second monitor has been placed directly to the right of your current screen. This second screen acts as an extension of the Windows desktop area, what I mean is that it is essentially blank space where you can position programs and windows on.

If you were to move the mouse to the right hand edge of your current screen now it would hit the edge of the screen and stop moving.

If we go back to our imaginary scenario with a second screen positioned directly to the right of your current one, then if you moved your mouse pointer to the edge of your first screen it would instantly appear on the left hand side of the second screen.

There would be no barrier between the two screens and the mouse would seamlessly move between both screens as you moved the pointer between them.

Moving on from this you can then 'drag' any program or window from your first screen on to the second screen. So if you are trading you could drag a chart from the main screen to the second screen and then open up another chart on the first screen.

You could see both charts at the same time, one on each screen, and they would both continue to update just as they would if you had them all on the same screen.

Once you understand this concept you can then see how beneficial it will be to have multiple screens available to you whilst trading.

Part 2 – How Traders Setup Their Trading Software On Multi-Screen Systems

How each trader specifically sets up their system is kind of a personal decision, I like my computer to operate a certain way, I like certain programs to show on certain screens, that being said…

After talking with literally hundreds of our trading customers about their computers and how they like to set things up, there are a few common traits which we have picked up on.

MT4 / MetaTrader 4

A lot of our customers use the MT4 platform, this is because a lot of the broker use it and this is because it has been built to be highly customisable.

Brokers can extend the performance of MT4 easily and provide it to their customers as a fully featured solution without having to engineer a full trading platform from scratch.

With MT4 you can open charts that are fed by feeds from your broker which display live financial data, you can then open up order windows and place trades instantly, it is a fully featured platform.

Running MT4 on one screen only gives you enough room for a handful of charts, if you like to see a lot of details then it may well be that you can only see one chart at a time.

It is a common theme that someone trading from just one screen is constantly flicking between charts as they can't see them all at once.

If you have multiple monitors attached to your PC you can improve the MT4 trading experience really easily.

The easiest way is to expand the MT4 program across all of your screens.

Imagine if you had 4 screens arranged in a square configuration, you can resize the main MT4 window and drag it across all 4 screens so that it spans them all. You can then open charts in MT4 and position them (drag them) to any position on any of the 4 screens. It is as easy as that.

As I mentioned before, MT4 is a highly customisable piece of software and many people have extended it. There are extensions that you can install which change how MT4 works when opening charts, they essentially open each chart as its own separate program or window.

You can then drag this chart window to anywhere on any screen whether that is inside the main MT4 program or not.

Traders with different sized screens or layouts that aren't square / rectangular can benefit from these types of extensions a lot.

Part 2 – How Traders Setup Their Trading Software On Multi-Screen Systems Continued...

Web Browser Based Platforms

Many brokers offer a web browser based solution to access charts and place buy / sell orders, i.e. you go to the brokers website and login to their online platform.

The benefit of these types of systems is that there is no need for any dedicated software installation or configuration, in theory you could login to your trading account on any internet connected computer in the world.

Different brokers web platforms have different features but they all kind of work in a similar manner to software like MT4, i.e. once logged in you can open charts, move them around in the web browser and resize them.

Some platforms allow you to open charts in new browser windows, for these you can simply move the windows around to any position on your screens as you see fit, if you've got this functionality from your broker then that's great.

Other web platforms are a bit more restrictive and won't let you drag a chart outside of the main browser area. For these we do a similar thing to MT4, you can make the browser window span multiple screens and then use the extra space provided to position your charts.

The CMC markets web platform works in exactly this manner.

Some brokers will let you login to their platform multiple times simultaneously, if yours will then you can open up your Internet browser and login to view charts as normal, you can then open a second internet browser and login again, you could then move the second browser to a different screen giving you the extra space to display more data.

TradeStation / NinjaTrader / Specialist Broker Platforms

Some of the bigger brokers have their own trading software platforms which they have developed in-house, examples of this include TradeStation and NinjaTrader amongst others.

Whether these programs support multi-screen computers and to what extent is very dependent on the individual platform, but as a generalisation, any software downloaded and run on a PC should be able to be expanded across multiple screens very easily.

TradeStation's platform calls the central workspace where you open up charts a 'desktop', their 'desktop' grants full functionality to all parts of the TradeStation platform.

For traders using multiple monitor configurations the platform allows you to open up multiple 'desktops', each of which can be positioned in its own screen, this provides full functionality on as many screens as you have connected.

NinjaTrader has 'workspaces', similar to the TradeStation desktop a 'workspace' allows access to a lots of charts and they can be dragged and resized across all of your connected screens.

Part 2 – How Traders Setup Their Trading Software On Multi-Screen Systems Continued...

Final Notes on Trading Platforms

As discussed above, pretty much all brokers will either have a web platform accessed through a browser, their own dedicated platform which is downloaded and run on your PC, or a customised version of MT4.

Each of these options offer compatibility with multi-screen setups, some more than others, but if seeing more than the odd one or two charts is going to help you become more successful as a trader, then going for a multi-screen setup is a purchase that could easily pay for itself very quickly.

Part 3 – How Many Screens Do You Need And What Size Should They Be?

This is another question we have been asked more than once and the answer is sometimes difficult to give because it really all depends on your specific trading style.

We do have some advice and pointers though which you can work through to hopefully give you a clearer idea of what might be the right number for you.

First of all, we will have a quick lesson on screen sizes and resolutions. Once you understand these you will then quickly be able to identify what kind of screen size will work best for you, and then how many of them you might need.

Screen Sizes & Resolutions

When choosing a screen size there are a few factors to think about, the first is the screen resolution and this is a really important concept to understand.

A screen resolution is the number of pixels that make up the image you see. A pixel is essentially a tiny dot which a monitor can change the colour of, when you look at a screen the image you see is made up of rows and columns of these tiny dots or pixels.

The size of the pixels and number that a screen can display determines how big things look on the screen and how much space you have to use.

A common screen resolution is 1920 x 1080, this means that across the horizontal row of the screen there are 1,920 pixels and it has 1,080 down its vertical column.

You could say the screen has 2,073,600 pixels (1920 x 1080) but screens are not typically listed with these kinds of counts.

Screen sizes are always listed in inches and this is a measurement across the diagonal of the usable screen (i.e. we do not measure any casing around the screen edge).

Common widescreen monitors these days are 21.5” and 24”, they are both popular sizes and often use a 1920 x 1080 resolution.

So, you could have a 21.5”, a 24” and a 27” widescreen all running at 1920 x 1080 resolutions, what would be the difference between the screens?

Well, obviously the 27” screen will be physically bigger than the other 2, but because it runs at the same resolution, the pixels which make up the image you see would have to be bigger.

Bigger pixels will make everything look bigger on the screen, so all things being equal the same line of text will appear bigger on a 27” screen than it would look on a 21.5” screen, even though there is technically no difference in size from the computers point of view.

You could also have 2 screens that are the same size, say 24” widescreens, but one has a resolution of 1920 x 1080 and the second has a ‘higher’ resolution of say 2560 x 1440.

Part 3 – How Many Screens Do You Need And What Size Should They Be? Continued...

When comparing these screens, the pixels on the higher resolution screen would have to be a lot smaller to fit in the same physical screen space, so going back to our line of text example, the text would appear a fair bit smaller on the higher resolution screen.

So, why would you want a higher resolution screen then if everything ends up being really small? The answer is that the extra pixels allow you to see more information on the screen at the same time, think of a spreadsheet like Microsoft Excel, more pixels means you'd see more rows and columns without needing to scroll.

Hopefully you can see that raising the resolution of your screens has benefits and drawbacks, you get more 'usable' space to layout your charts and programs but everything gets a bit smaller.

There is then a way to have your cake and eat it with higher resolution screens, well sort of.

Going for a higher resolution and a physically bigger screen means that you get that extra usable space benefit, and the physically bigger screen makes the pixels bigger, so that everything is still visible without needing a magnifying glass to see it all.

What is the drawback of going down this route? Basically it is the cost.

Larger and higher resolution screens cost considerably more than smaller lower resolution screens. Once you get past a certain level of resolution you also start needing some serious graphics card power to run them, which can also have a significant impact on your set up costs.

High Resolutions and Scaling

One final note on super high resolution screens is a Windows issue and it is to do with something called scaling. Without going too much into it here, when you run Windows (or any software really) on a really high resolution screen, then the pixels which make up the picture obviously end up being really small.

This can cause problems with programs. Let's say that your charting platform developers (the guys who designed it) decided that the button to place a new buy / sell order would be 200 pixels wide by 100 pixels tall. It may look perfect on a 1920 x 1080 resolution screen (or anything using similar sized pixels).

If you view this program on a super high resolution screen such as a new 4K monitor (which incidentally have resolutions of 3840 x 2160) then the same 200 x 100 pixel button is going to be a quarter of the size it should be, which may make it almost unusable if not very difficult to see and click in a hurry.

This scenario I've described is actually a pretty big problem with high res screens, to try and counter it Windows lets you scale the interface to a higher percentage.

For example if you scaled Windows to 150% then it would resize everything you see on the screen to make it 150% of its originally designed size.

Part 3 – How Many Screens Do You Need And What Size Should They Be? Continued...

In theory this should be a problem solver, in reality some programs scale really well, some scale badly (think blocky / fuzzy graphics) and some programs refuse to scale at all.

Windows 10 improved the problem somewhat but there are still many programs around that do not scale well at all. It's not just traders and multi-screen users that suffer due to this, it affects pretty much anyone using really higher resolution screens.

Working Out Your Needs

So, hopefully you've survived my crash course in screen sizes and resolutions? Now we need to put this new found information to some use.

The best way I can recommend deciding on screen resolutions and numbers is to first take a look at your existing screen and then work on from there.

If you don't know the size of your current screen it may be written on the back, you could try googling the make and model number, or you could get a tape measure out (remember to measure the diagonal).

Once you have this you need to work out your resolution, this is easy to do. In Windows 7 / 8 simply right click on a blank portion of your desktop screen and select 'Screen Resolution' from the menu, in Windows 10 this has changed to 'Display Settings'.

The windows display settings screen will open and there will be an image of a small screen which represents your monitor, underneath are some options and settings, one of these dropdown lists will be resolution and handily it will show you your resolution.

(In Windows 10 you have to hit the 'Advanced display settings' link before you can see the resolution setting for your screen).

Now you should have your screen size and resolution.

With this information in hand we can now work out the rough size of each pixel on both your current and new screens, using this we can determine whether programs and text are going to be massively affected by a switch to a different screen size / resolution. To work this out follow this short process.

Let's say you are trading on a laptop with a 14" widescreen running at a resolution of 1440 x 900, this screen would likely be around 300mm wide (you can measure it to be more precise).

We can work out the rough width of each pixel by dividing the width of the screen by the number of horizontal pixels, this would be 300mm / 1440 pixels which gives a rough size of 0.208mm per pixel.

Moving up to a 24" widescreen will give you an extra 10" of diagonal space which will work out to somewhere around 230mm of actual extra screen width (540mm is the total width of a standard 24" screen).

Pro Tip: Most new monitor specs list the diagonal size of a screen as well as the width / height of the physical screen size.

If your new 24" widescreen had a resolution of 2560 x 1440 then a bit of

Part 3 – How Many Screens Do You Need And What Size Should They Be? Continued...

math shows us that each pixel is roughly 0.211mm wide (540mm width / 2560 pixels).

Essentially the pixel size would physically remain almost identical between the two screens so text, charts and programs will look the same size on both screens, you will just get a lot more space on the 24" screen to display more data (especially if you have a few of them on your new system).

If you work out the pixel sizes and there is a bigger difference (i.e. the pixels are getting a lot smaller) then you may need to make charts bigger on the new screens to be able to see the details or even scale the interface which is going to reduce the amount of space that you will be left with.

Finally, you need to run your charting / trading platform and start opening up charts on your current screen. Resize them to a size that you are comfortable with making sure you are seeing as much detail as you would need to see to successful read the data as it comes in.

How many charts can you fit on your screen at the same time?

Judging this and a bit more math will tell you how many pixels you will need to see everything you want to see, and then you can tell how many screens you will need to spread this across.

If your current screen is a 21.5" monitor running at 1920 x 1080 and you are happy with two charts being displayed at the same time, and you need to see 6 charts to avoid having to flip between them, then a triple monitor system with three 1920 x 1080 monitors would give you enough room to lay everything out that you need.

Final Considerations On Screen Sizes & Numbers

Do you need a screen dedicated to executing orders as well as just viewing charts?

Do you browse the Internet / watch films / stream Bloomberg whilst trading? Many traders do this, if that sounds like you then are you going to dedicate a screen to this kind of activity as well as your charting screens?

In terms of computers the number of screens they can support tends to go up in sets of 2, this means that moving from a 5 to a 6 screen setup will generally only cost you the price of an extra screen. Take this into account when planning your setup, everyone can always find something to put in that extra screen.

Going for a lot of high resolution screens is going to put up the cost of the screens quite a lot, the trading computer will also require more power which could also significantly increase the cost of your new setup.

Try to think about your physical environment when considering screens sizes and numbers. Some people assume 21.5" widescreens will be too small, but if you have six of them on a desk will they still feel too small?

Likewise, you might decide that some slightly higher resolution 27" widescreens gives you the perfect amount of space to use, but are you going to be comfortable seated in front of 4 of them for long periods of time?

Part 3 – How Many Screens Do You Need And What Size Should They Be? Continued...

Screen sizes are a personal choice, I've experimented with lots of different sizes and I find that having more than one 27" widescreen on my desk is too much for me.

I can deal with 21.5" / 24" screens and often have four of these in front of me, but switching to just two 27" widescreens felt like too much screen too close to me.

You can run different sized screens at different resolutions, there is no problem doing this as far as the computer goes, the only thing to consider is that if there is going to be a big difference in resolution between your screens then you may run into scaling issues.

The only other thing to consider when running different sized screens is getting them physically lined up together, for some layouts this is not a problem, for others you could end up with gaps between the screens or different heights, this is more of an aesthetic consideration though and will not bother some people.

Thank You For Downloading & Reading This Guide

Thank you for downloading and reading this guide to using Multiple Monitors to power your trading success, I hope you found it useful?

If you are ready to move forward with a new multi-screen trading system then we are ready to help you put it together.

We have a range of multi-screen computers which are all very well suited to traders like you, we even have a dedicated Trader PC.
www.multiplemonitors.co.uk/computers/

In terms of stands we have our own exclusive Synergy Stand range which can hold up to 28” widescreens in pretty much any layout you can think of.
www.multiplemonitors.co.uk/stands/

We also offer a range of monitors which we know are a good fit for multi-screen usage.
www.multiplemonitors.co.uk/monitors/

Our bundles are popular with customers as they offer everything you need to get instantly up and running in one easy package. You can customise your own bundle with your perfect choice of stand, screens and PC and we will supply it including all the required cables and any required screen adapters.

Bundles also come with money saving discounts, free (UK) delivery and free long length high quality cables.
www.multiplemonitors.co.uk/bundles/

If there is anything else we can help you with just let us know.